

Seasonal Drought Prediction in Africa Utilizing a Combined Statistical and ML Approach

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1 Introcutiion: Hybrid approaches to seasonal drought prediction

Drought is one of the meteorological extreme events that has the greatest impact on agricultural, social, and economic conditions. Extreme droughts occur in almost all climate zones and vary widely in duration, intensity, and regional extent. A distinction can be made between meteorological, agricultural, and socioeconomic droughts. Agricultural and socioeconomic definitions focus on the impacts of meteorological droughts on society. However, the prediction of meteorological drought events even with state-of-the-art global climate models (GCMs) often lacks significant prediction skill. On seasonal time scales, GCMs are mostly used to predict precipitation and temperatures, which are consequently used to calculate drought indices, such as the Standardised Precipitation Evaporation Index (SPEI). Despite huge advancements in seasonal climate predictions in the last decades, climate models still possess significant biases and uncertainties when predicting drought events (Wu u. a., 2022). Especially after the loss of initialization effects, GCMs still show significant errors in their dynamical representation of meteorological droughts (Wu u. a., 2022). In alternative approaches, statistical methods can be used to directly predict the occurrence and intensity of drought events from historical data. However, also statistical models, such as Machine Learning (ML) or regression methods, are limited in their capability to represent nonlinear dynamics reliably (Hao u. a., 2018). While being able to learn and reproduce non-linear dynamics from its training data, ML lacks interpretability and a good representation of overall distributions. In order to combine the strengths of both methods, hybrid approaches based on a mix of GCM prediction and ML postprocessing have recently gotten a lot of attention (Li u. a., 2021; Wu u. a., 2022). Due to the sparse amount of training data and a focus on reliability and explainability, these hybrid approaches often use relatively simple ML models. In comparison to other fields in meteorological or climate research, modern ML techniques like deep neural networks or transformer models seem to be less suited than reliable and cost-efficient solutions like random forest models (Thomasson, 2023; Feng u. a., 2024; Karsisto, 2025). Therefore, this work will also focus on the use of random forest or as an extension boosted decision trees for the postprocessing step. Especially extreme gradient boosting (XGBoost) has shown to be a reliable, efficient and flexible option for forecast postprocessing (Feng u. a., 2024). However, XGBoost has shown to sometimes struggle with extreme outliers and to correct for systematic biases in the variance and distribution of its input data (Feng u. a., 2024). Thus, an intermediate statistical step will be performed. In this step, quantile mapping (QM) matches the original hindcasts distribution to that of the reference data. Since QM cannot by design correct the trend or time evolution of its input data, the combination of QM and XGBoost should be able to account for each others weaknesses. Thus I hope with the framework of QM plus a ML prediction of the remaining residuals to reach an overall significant improvement of the original prediction skill.

Despite the success of ML postprocessing in some fields of climate science, they have not yet been successfully applied to seasonal predictions. Especially the prediction of extreme events, such as droughts,

has only recently gotten more attention in seasonal prediction systems. Thus, there is still significant room for improvement, especially in the postprocessing of these predictions. The DWD for example does not possess an operational seasonal forecast postprocessing including ML techniques to this day. This work tries to close this gap. By combining dynamical predictions of climatic variables on seasonal timescales with a statistical postprocessing step, I will analyse if ML models can be used in order to improve seasonal drought prediction. There are two possibilities to improve drought prediction. Either one can deterministically postprocess an entire prediction of a chosen drought indicator. For this work the SPEI is chosen as a metric for drought severity, including both precipitation and potential evapotranspiration (PET) in its calculation. In this way the postprocessing step aims at improving the overall forecast skill of the SPEI, without specifically targeting the occurrence of drought events. Alternatively, the occurrence of individual extreme drought events can be first extracted from the chosen drought index. Consequently, the ML step can focus on just improving the forecast of these extreme events by either predicting their intensity or by probabilistically predicting the absolute occurrence of a drought event above a certain threshold. In this work both options will be tried out and evaluated in regard to their potential for improvement of prediction quality.

In order to come to an informed conclusion, first an overview of the utilized methods and data will be given in chapter . Both the original dynamical hindcasts and the observational data used for the ML training will be described in detail in section 2.1. The used ML framework as well as additional post- and preprocessing steps will be documented in 2.2. Afterwards, the results of the subsequent analysis will be shown in chapter . At last, the results will be discussed and a conclusion will be drawn in chapter in regard to this hybrid approach's ability to significantly improve the seasonal prediction of drought events.

2 Data and methods

2.1 Data

This work uses the climatic waterbalance (CWB) defined as:

$$CWB = P - PET, \quad (2.1)$$

with P as the total monthly precipitation and PET the potential evapotranspiration. Due to the limitation in output variables from the DWD seasonal prediction system, a relatively simple version of the PET is used, only taking monthly mean temperature as an input value. For the training and evaluation of the network, three datasets are used and therefore have to be discussed. On the one hand, the hindcasts used for the ML training and therefore the input into the postprocessing workflow are the seasonal climate hindcasts by the German Weather Service (DWD). On the other hand, a mixture of 2 reanalysis datasets based on remote sensing observations have been chosen as reference data and therefore assumed ground truth. This reference CWB is build from the TAMSAT precipi-

tation reanalysis and the ERA5 reanalysis of near-surface air temperature. In the following section, these datasets, their strengths and limitations as well as current use cases in research will be discussed.

2.1.1 The GCFS Seasonal Forecast System

Since 2016, the DWD is operationally running the current GCFS forecast system. It provides seasonal forecasts up to six months ahead using a fully coupled Earth system model, including atmosphere, ocean, land, and sea ice component (Fröhlich u. a., 2021). GCFS2.0 is based on the Max-Planck Insitutes high resolution earth system model (MPI-ESM-HR), with an atmospheric resolution of approximately 70 km at 50°N and an ocean resolution of approximately 0.4°, representing a major upgrade from GCFS1.0 (Fröhlich u. a., 2021). Initial conditions are generated using continuous nudging assimilation toward ERA-Interim atmospheric reanalysis combined with ORAS5 ocean reanalysis, as constructed in (Baehr u. a., 2015). Ensemble forecasts are produced using perturbations in atmospheric parameters and ocean bred vectors, with 30 hindcast members. Of these 30 members, 25 are currently available for download from the copernicus data store and used for the training of the ML algorithm. Compared to the previous version, GCFS2.0 shows significantly improved winter forecast skill, particularly for Northern Hemisphere circulation patterns, where NAO prediction skill approximately doubled compared to GCFS1.0 (Fröhlich u. a., 2021). However, weaknesses remain, including reduced skill in the tropical Pacific and Atlantic, which negatively affects ENSO prediction. In regard to the study region of this work, a cold bias over northern Africa and a warm bias over southern Africa is still apparent, even if reduced compared to GCFS1.0 (Fröhlich u. a., 2021). GCFS2.0 also exhibits a persistent tropical warm bias, likely related to atmospheric parameterization changes. Overall, GCFS2.0 improves winter predictability and regional forecast skill, but limitations remain in tropical processes and seasonal-dependent performance (Fröhlich u. a., 2021).

The TAMSAT precipitation dataset

The TAMSAT African Rainfall Climatology and Time Series (TARCAT) is a 30-year precipitation dataset (1983–2012) developed to provide temporally consistent rainfall estimates across Africa (Maidment u. a., 2020)¹. TARCAT is based on Meteosat thermal infrared (TIR) satellite imagery calibrated against rain-gauge observations to estimate rainfall across data-sparse regions. The dataset is produced at 0.0375° (ca. 4 km) spatial resolution and 10-day (dekadal) temporal resolution, enabling regional-scale climate and drought analysis. Rainfall estimates are derived using cold cloud duration (CCD) from TIR imagery, based on the assumption that cold cloud tops are associated with deep convective rainfall. The TAMSAT algorithm applies fixed monthly calibration parameters derived from gauge data, ensuring temporal consistency without requiring contemporaneous observations. This calibration approach reduces biases caused by declining gauge networks and enables reliable long-term rainfall variability analysis. TARCAT successfully reproduces spatial rainfall patterns and interannual variability, with correlations up to 0.85 with CRU rainfall data. However, the dataset shows a dry bias of approximately 20% due to its optimization for drought monitoring. Overall, TAMSAT provides high-resolution rainfall estimates particularly valuable in gauge-sparse regions of Africa, though limitations remain due to satellite calibration uncertainties and convective-based rainfall assumptions.

¹All information on the TAMSAT precipitation dataset in this subsection are taken from (Maidment u. a., 2020), unless explicitly marked differently.

The TAMSAT TARGAT dataset addresses sparse gauge observations by using fixed climatological calibration parameters derived from historical rain-gauge data rather than merging contemporaneous observations. This approach avoids biases introduced by changing gauge coverage over time. Rainfall estimates are then generated solely from satellite-derived cold cloud duration (CCD) fields, allowing spatially complete precipitation estimates even where no ground observations exist. For missing satellite observations, gaps of up to 6 hours are statistically interpolated, while longer gaps are supported by climatological 10 day averages. Due to its calibration for regional droughts in Africa, the TAMSAT reanalysis presents an ideal baseline for this work's postprocessing of seasonal drought predictions.

2.1.2 The ERA5 reanalysis temperature dataset

ERA5 is the fifth-generation global atmospheric reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) and extended to cover the period 1940–2022 (Soci *u. a.*, 2024)². The dataset provides hourly global atmospheric, land surface, and ocean wave fields at approximately 31 km horizontal resolution. ERA5 fills observational gaps using data assimilation within a global numerical weather prediction model, combining observations with physically consistent model forecasts. Before the satellite era, particularly in the Southern Hemisphere, ERA5 fields were therefore largely statistically constrained by the model, reflecting greater uncertainty due to limited observations. ERA5 assimilates sub-daily in-situ and satellite observations, increasing from about 17,000 observations per day in 1940 to 25 million per day by 2022. As a result, the quality of ERA5 improves over time, especially since the introduction of satellite observations in the 1970s. Surface air temperature at 2-m height is one of the key variables of the ERA5 reanalysis. ERA5 shows cold biases over land during early decades (1940–1957) due to limited observational constraints. After the late 1950s, ERA5 temperature trends agree well with independent observational datasets such as Berkeley Earth and GISTEMP. Despite the early-period limitations, ERA5 provides one of the longest and most consistent global temperature reanalyses.

2.1.3 Data Preprocessing

Before hindcast and observational datasets can be fed as input to the ML algorithm, they have to be preprocessed. First, in order to ensure parallel resolutions, both observational datasets are remapped to share the common resolution of the GCFS hindcasts. Since the GCFS precipitation estimate is measured in total precipitation rate, it also first has to be converted to mean precipitation per month to match the TAMSAT dataset via

$$P = tprate \cdot spm, \quad (2.2)$$

where *spm* is the amount of seconds per month.

Afterwards, the potential evapotranspiration (PET) has to be calculated from the two temperature estimates. Due to the lack of other variables, the Hamon method for the calculation of the PET is utilized, only taking the daily mean temperature and latitude at that gridcell as inputs. For lack of

²All information on the ERA5 temperature dataset in this subsection are taken from (Soci *u. a.*, 2024), unless explicitly marked differently.

daily forecast data, the PET is assumed to be stationary over the whole month at the monthly mean temperature. The Hamon method calculates PET as

$$PET = c \cdot \frac{N}{12} \cdot P_t. \quad (2.3)$$

In this equation, c is a constant coefficient, N denotes the number of daylight hours and P_t is the saturated water vapor density at the daily mean temperature.

Then, the climatic water balance (CWB) is calculated as the difference between daily precipitation and PET (see equation 2.1). Finally, a monthly mean of the CWB is taken to return to seasonal timescales.

2.2 Methods

This work will use two distinct steps for the postprocessing of seasonal climate predictions. First, statistical quantile mapping will reduce the distributional error and biases of the original hindcast as far as possible. Subsequently, an ML algorithm will postprocess the residuals left over from the quantile mapping step. In the following, both the quantile mapping and the ML approach will be described in detail.

2.2.1 Statistical Quantile Mapping

Quantile mapping (QM) is a statistical bias-correction method that can adjust a statistical distribution to match another. QM corrects systematic biases by redistributing the values of an input variable to the empirical cumulative distribution functions (CDFs) of target data. Because the full distribution is adjusted, quantile mapping can correct biases in mean, variance, and extremes (Schwertfeger, 2023).

For this work, the Python Cmethods implementation is used (Schwertfeger, 2023), where quantile mapping is defined by mapping values from a modeled control distribution to the observational reference distribution. The package was especially constructed for the use on climate data. The corrected value is calculated via a series of correction steps. First, the quantile value of time step i saved as $\epsilon(i)$.

$$\epsilon(i) = F_{sim,p}[X_{sim,p}(i)], \epsilon(i) \in \{0, 1\}, \quad (2.4)$$

where $F_{sim,p}$ represents the cumulative distribution function of the input (model) data of the variable value $X_{sim,p}$ for a certain quantile p . In the next step, the quantile value is inserted into the inverse cumulative distribution function of the reference (observational) data $F_{obs,h}^{-1}$. This yields the corrected quantile mapped value $X_{sim,p}^{QDM(1)}$:

$$X_{sim,p}^{QDM(1)}(i) = F_{obs,h}^{-1}[\epsilon(i)]. \quad (2.5)$$

Therefore, QM overall determines the quantile value of the reference data for the corresponding quantile of the input data (Schwertfeger, 2023).

Despite its flexibility, QM has several general limitations. First, QM assumes that model biases estimated during the calibration period remain stationary in the prediction period, which may not hold under changing climate conditions (Cannon *u. a.*, 2015). However, with the prediction on seasonal timescales, forecast data can be assumed to be within the training period’s general distribution. Nonetheless, detrending training and evaluation data can be introduced to counter this out-of-distribution problem under changing climate conditions. Furthermore, quantile mapping modifies the marginal distribution only and does not correct temporal sequencing, persistence, or physical process representation in climate models. In order to account for these weaknesses, the correction of temporal sequencing and physical representations is then tackled by the ML step afterwards.

Quantile mapping is widely used for climate hindcast post-processing because climate models often exhibit systematic biases in both temperature and precipitation. QM methods have been shown to substantially improve agreement between modeled and observed hydro-climatic variables and reduce systematic errors in climate simulations (Enayati *u. a.*, 2021). A key strength of QM for hindcast correction is that it adjusts the entire distribution rather than only mean values, improving representation of extreme events as well as mean and variability representations. However, quantile mapping may also alter modeled climate signals. For example, Cannon *u. a.*, 2015 show that quantile mapping can modify projected trends in extremes even when mean trends are preserved, highlighting trade-offs between bias correction and trend preservation.

Overall, quantile mapping provides a robust method for post-processing climate hindcasts. Nonetheless, fundamental limitations especially in the correction of temporal sequencing remain that have to be addressed in addition to the quantile mapping. This is where this work employs a second postprocessing step, utilizing ML to address the remaining uncertainties in the prediction.

2.2.2 Residual correction with mixed effect random forests

In a second postprocessing step, I try to predict the remaining differences between the QM results and the observational reference data. In order to reach a stable, cost-efficient and at the same time sophisticated ML setup, this work relies on a mixed effect random forest (MERF) architecture. This is based on a relatively simple gradient boosted Random Forest model and supplemented by a grouped random effect. Therefore, it is important to first explain RFs and the expanded gradient boosted version (XGBoost) before then explaining MERFs and the implications for this work.

Random Forest Machine Learning Models

In general, Machine Learning can be defined as algorithms that can learn from data without being specifically programmed (Louppe, 2014). An algorithm learns from data when it improves at a certain task, mostly regression or classification tasks, with its input data. One of the first subsets of Machine Learning tree structured models. Basically, tree structured models divide the input spaces of a classification or regression task into subspaces (trees) and subsequently assign prediction values to all objects within a subspace. This division is often conducted in different steps resulting in internal trees (subspaces that are divided again) and terminal trees (the final subspaces, also called leafs) (Louppe, 2014).

In supervised ML, the construction of decision trees is then driven by finding a division, which minimizes a loss for a set of training data. Regression or prediction problems are in this case defined

as learning a target value (output) for each predictor (input). The loss for one tree ϕ can be defined as the combined loss for each individual terminal tree t with an element y_t :

$$Err_{train}(\phi) = \sum_{t \in \tilde{\phi}} p(t) \cdot (1 - p(\tilde{y}_t|t)), \quad (2.6)$$

where Err_{train} is the decision trees total error on the training dataset and $\tilde{\phi}$ denotes the total set of terminal trees of the overall tree ϕ (Louppe, 2014). Mathematically, it can be quite easily proven that each new terminal tree reduces the training error until the tree is finally saturated and has an individual leaf for each predictor sample. However, such an overcomplex model might reduce the training error for a model, but in most cases then loses its ability to generalize its results for out-of-distribution validation data, which nearly all prediction tasks include. Thus, metrics usually have to be chosen to stop the development of additional leafs during training, which minimize the error not only on the training dataset, but more importantly on a separate validation dataset. Additionally, further stopping criteria, like restricting the amount of leafs per tree as well as the overall tree’s maximum depth, can reduce the risk of overfitting to training data.

The basic idea behind Random Forests (RF) as decision-based ML methods is to introduce an ensemble of randomly perturbed decision trees, which are at the end combined again in order to reach a more stable prediction. Usually, this combination is conducted by taking the average prediction of all individual trees as the average prediction is assumed to minimize the error compared to the individual, more variante predictions. This assumption can be made, as an ensemble is less likely to get stuck in local loss minima and is able to better capture generalization possibilities to unrelated validation data. Since the 1990s random forests have been developed and mostly changed in the manner, in which the random perturbations are introduced to the ensemble (Louppe, 2014). Since then, RFs gained widespread use especially due to their quality at predicting out-of-bag validation data. Additionally, they possess the straightforward ability to assess the importance of individual predictors for the prediction quality. This feature of decision trees adds a lot of interpretability and reliability to RFs compared to black-box models, such as neural networks. Similarly, due to their ensemble nature, RFs allow the estimation of the proximity of predictor values in the RF’s feature space by determining the co-occurrence in a common leaf over an ensemble of decision trees. In this way, RF results can again be explained better, while outliers in the training dataset become easily identifiable. Finally, compared to other more complex ML methods, RFs show high consistency at low computational costs and a relatively simple algorithmic setup. Therefore, they require less hyperparameter tuning as well as little domain specific ML knowledge for implementation (Louppe, 2014).

Due to all these advantages, RFs have found many applications in the climate sciences of recent years (Das u. a., 2022; Valecha u. a., 2018). They even have already been combined with quantile mapping in order to improve bias correction of climate projections (Raeesi u. a., 2024). In their study, Raeesi u. a., 2024 have found that RFs can significantly reduce projection errors even after the quantile mapping step. However, these methods have neither been utilized for seasonal predictions yet, nor for the consideration of extreme events. Additionally, RFs can run into difficulties when tasked to predict complex patterns. Therefore, it is worth looking at further developments of decision tree based ML methods in order to gain maximum performance from limited training data.

Gradient Boosting with XGBoost

XGBoost is a type of gradient boosting algorithm that builds an ensemble of decision trees sequentially, where each new tree focuses on correcting the errors made by the previous ones. Unlike random forests, which aggregate the predictions of many independent trees, gradient boosting improves prediction accuracy by iteratively minimizing error. (Karsisto, 2025) On a general level, XGBoost remains a decision tree based algorithm, where the final prediction can be expressed with

$$\tilde{y}_i = \sum_{k=1}^K f_k(x_i), \quad (2.7)$$

where y_i is the prediction value of the model for the feature vector x_i at sample i and f_k is the leaf score for the k th tree. Compared to an RF ensemble, each tree is iteratively trained on the residual of the last prediction. Therefore, it is able to learn from the results of the previous trees until a maximum number of K trees is obtained.

These changes in algorithm lead to generally accepted improved results. For example, XGboost algorithms have shown to be more accurate, faster and less prone to tree overfitting compared to regular random forest architectures (Chen und Guestrin, 2016). However, they have also shown to sometimes miss nonlinear effects that more complex ML methods, like deep neural networks are able to extract. Therefore, a challenge remains in finding a method that retains the stability and computational efficiency of decision tree based methods, the improved learning of gradient boosting over random forests and can capture more complex nonlinear effects.

Adding Group Specific Effects with Mixed Effect Random Forests

One idea to improve on ensemble tree methods is the idea of mixed-effects regression trees. In the basic idea of this algorithm developed by Hajjem u. a., 2011, fixed and random group effects are dissociated and corrected separately. This can be expressed as

$$y_i = X_i\beta + Z_i\text{plus}\eta_i, \quad (2.8)$$

where y_i is the model prediction, X_i is the matrix of fixed-effect covariates, Z_i is the matrix of random-effects covariates and η_i denotes the remaining error. For a decision tree, or an ensemble of trees (Hajjem u. a., 2014) to predict fixed relationships in the data, the random-effect of group clustering is estimated at each iteration and removed from the data. The remaining cluster-specific component can then be estimated individually, for example with a best linear unbiased predictor (BLUP). So basically, the algorithm first removes the random effect from the predictor at each iteration. Afterwards, the prediction model, often an ensemble based decision tree model, such as RFs or XGBoost, estimates the fixed effects. Then the clustered random effect is newly estimated from the remaining residuals and again subtracted from the original predictor for the next basemodel iteration (Hajjem u. a., 2011).

Simulations show that mixed effect random forest (MERF) methods show significant improvements compared to standard ensemble decision tree algorithms when group effects are non-negligible (Hajjem

u. a., 2014). The improvements come especially in the estimation of non-linear relationships in the prediction data. Therefore, MERFs should represent a viable option for the postprocessing of climate predictions, where data can be grouped by region or physical properties, for example its value in a certain climatic variable. In this work this mixed effect approach will be combined with an XGBoost base model in order to profit from the fast and accurate learning of XGBoost models while at the same time using the mixed effect method for improving on the remaining nonlinear residuals.

3 Results

3.1 Setup

Section 2.2 describes the many advantages that mixed effect boosted forests can bring as a method to the postprocessing of clustered climate data. Now this chapter brings the methodological concepts from section 2.2 together with the available data (section 2.1) in order to outline the step-by-step approach to the performed analysis.

At first, all data is preprocessed according to the steps described in section 2.1.3 in order to gain a *CWB* dataset for both reference and hindcast data. Then, the preprocessed hindcast data is quantile mapped to fit the reference data's distribution. Afterwards, a mixed effect ensemble tree method uses the quantile mapped hindcasts as training predictors in order to predict its residuals to the reference data *CWB* as a target variable. This mixed effect model utilizes an XGBoost base with a BLUP estimation of the grouped random effects. In addition to the quantile map hindcasts, the XGBoost base model is given several other predictors, including reference climatology, reference data in the three months before the hindcast start, regional information (latitudes and longitudes of each grid cell) and reference variability of the climatological period. The impact of each predictor will be analysed and the choice of predictors will be accordingly adjusted in order to gain the best possible prediction. Then, the choice of clustering will be analysed with the most important clustering choices being the grouping by quantiles of the most important predictors, by grid cell or by region both physically and in the predictor feature space. The analysis will also include a short comparison to simpler ML models, like basic Random Forest predictors and XGBoost models without a mixed effect implementation. For each year only one hindcast is extracted in order for the network to be able to learn leadmonth and calendar month specific dynamics in the quantile mapped residuals. Each hindcast covers the 6 month time period from June to November, which corresponds to the dry season of especially southern and south-eastern Africa. Of the 42 available years from 1983 to 2025, 4 years will be laid aside from the training procedure for both validation and testing. Finally, all postprocessing models will be evaluated according to their effect both on deterministic metrics such as RMSE or MAE and in regard to their probabilistic prediction of extreme events. Afterwards, the results of this analysis will be discussed in section 4.1.

3.2 Quantile Mapping

The QM algorithm, described in section 2.2.1, refittes the original GCFS Hindcast *CWB* distributions (section 2.1.1) to fit the reference observational *CWB*, calculated from the Tamsat precipitation (section 2.1.1) and the ERA5 temperature dataset (section 2.1.2). The main variable in statistical QM that can be varied is the amount of quantiles. This value has to be fixed at the beginning of the QM step in order to sort the hindcast and reference distribution values into the corresponding quantile groups. In this consideration, a larger amount of quantiles generally leads to a closer fitting of the distributions, while it also comes hand in hand with higher computational costs. There is a limit to the accuracy of quantile mapping, which is approached exponentially. So in order to get an amount of quantiles that stays computationally efficient while retaining close to maximal accuracy a number of different values for the amount of quantiles n_q has to be tested.

To evaluate the quality of the utilized QM step, both deterministic and probabilistic metrics are chosen in order to analyse the effect of the QM step on the overall as well as the extreme event prediction.

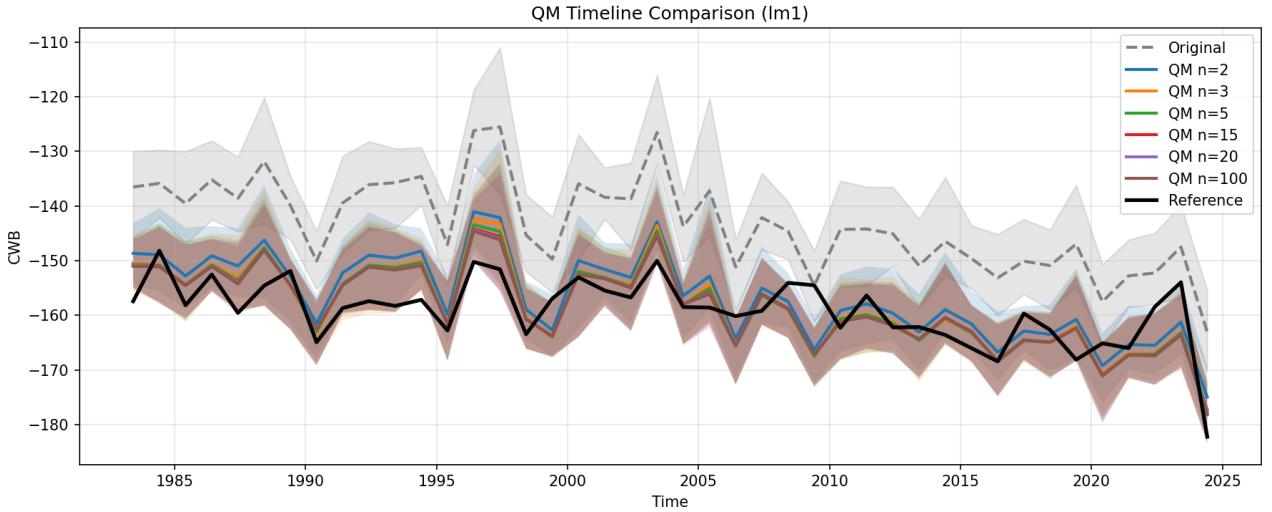


Fig.3.1 | Timeseries comparison of quantile mapping implementations. Quantile mapped hindcast ensemble means *CWB* with min/max spread compared to the original GCFS hindcast and reference *CWB* for the first leadmonth. The differenceThe original hindcast timeseries is displayed as dotted line, while the reference *CWB* is shown in black.

Figure Fig. 3.1 illustrates two very important results. On the one hand, it is visible that the QM procedure significantly reduces the mean bias in the original hindcast's estimate. Although it does not correct for temporal trends in interannual variability, it completely erases the mean bias over the entire 40 year period. On the other hand, figure Fig. 3.1 illustrates that the amount of quantiles has only very little impact on the quality of the QM estimate. Just by visual inspection the timeseries become indistinguishable at $n_{quantiles} > 15$. The same results can be established from a comparison of the *CWB* distributions.

Again, a minimal reduction is still visible up to a value of $n_{quantiles} = 15$. The illustration also shows that the quantile mapping does not only shift the mean distributional bias but also corrects for some distributional mismatches. However, it also shows the limits of the QM procedure, which cannot perfectly match the distributions towards the reference data even at very high amounts of quantiles. Further inspection still shows a minimal reduction in root mean squared error (RMSE) for all leadmonths up to a value of $n_{quantiles} = 20$. Therefore, $n_{quantiles} = 20$ is chosen in order to provide

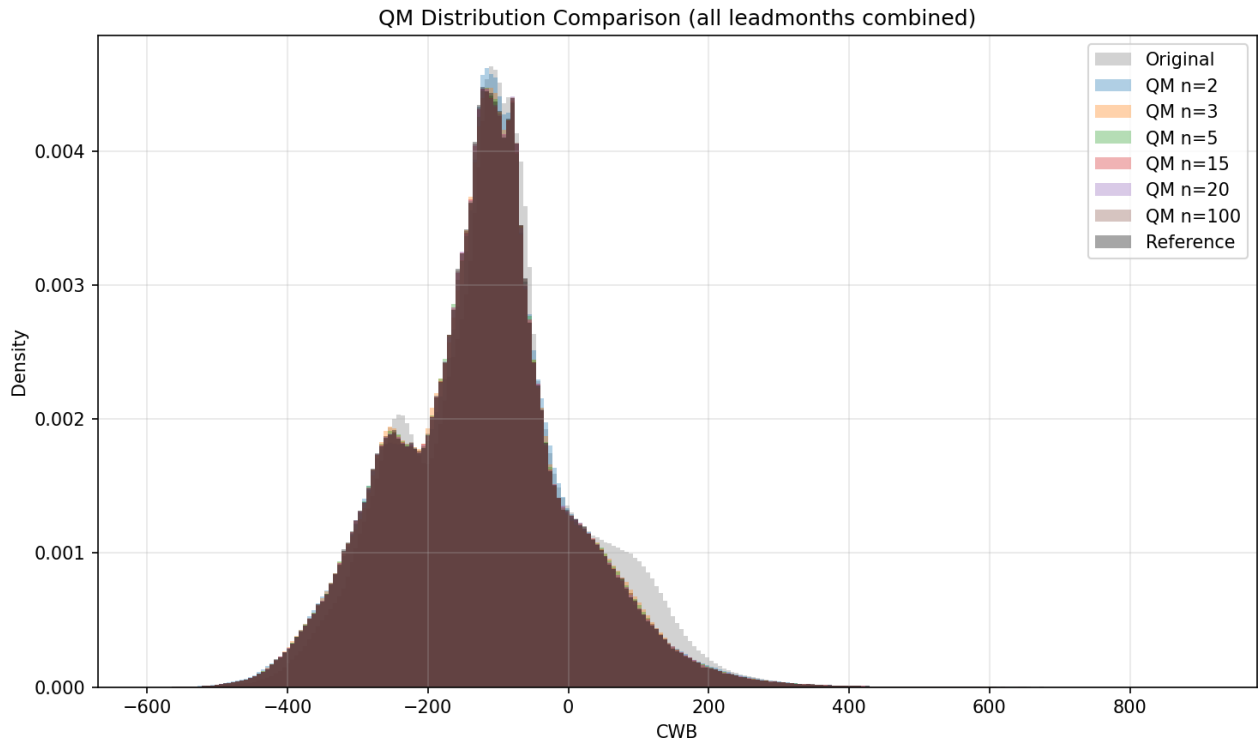


Fig.3.2 | *CWB* distribution comparison of quantile mapping implementations. Quantile mapped hindcast ensemble mean *CWB* distribution compared to the original GCFS hindcast (grey) and reference *CWB* (black) for the first leadmonth.

the best possible input to the ML postprocessing step, while reducing the computational costs as much as possible.

3.3 Deterministic Prediction Improvement

3.4 Probabilistic Prediction Improvement

4 Discussion and Conclusion

4.1 Discussion

4.2 Conclusion

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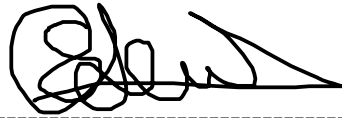
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Erklärung über Selbständiges Verfassen

Ich versichere, dass ich die vorstehende Arbeit selbständig und ohne fremde Hilfe angefertigt und mich anderer als der im beigefügten Verzeichnis angegebenen Hilfsmittel nicht bedient habe. Alle Stellen, die wörtlich oder sinngemäß aus Veröffentlichungen entnommen wurden, sind als solche kenntlich gemacht.

Hamburg, 25.09.2025

Datum, Ort

A handwritten signature in black ink, consisting of a large, stylized 'S' followed by a series of loops and a long horizontal stroke extending to the right.

Unterschrift